

Ensemble Classification and IoT-Based Pattern Recognition for Crop Disease Monitoring System

Gayathri Nagasubramanian¹, Rakesh Kumar Sakthivel², Rizwan Patan³, *Member, IEEE*,
Muthuramalingam Sankayya, Mahmoud Daneshmand⁴, *Life Senior Member, IEEE*,
and Amir H. Gandomi⁵, *Senior Member, IEEE*

Abstract—Internet of Things (IoT) in the agriculture field provides crops-oriented data sharing and automatic farming solutions under single network coverage. The components of IoT collect the observable data from different plants at different points. The data gathered through IoT components, such as sensors and cameras, can be used to be manipulated for a better farming-oriented decision-making process. This work proposes a system that observes the crops' growth and leaf diseases continuously for advising farmers in need. To provide analytical statistics on plant growth and disease patterns, the proposed framework uses machine learning (ML) techniques, such as support vector machine (SVM) and convolutional neural network (CNN). This framework produces efficient crop condition notifications to terminal IoT components which are assisting in irrigation, nutrition planning, and environmental compliance related to the farming lands. In this regard, this work proposes ensemble classification and pattern recognition for crop monitoring system (ECPRC) to identify plant diseases at the early stages. The proposed ECPRC uses ensemble nonlinear SVM (ENSVM) for detecting leaf and crop diseases. In addition, this work performs comparative analysis between various ML techniques, such as SVM, CNN, naïve Bayes, and *K*-nearest neighbors. In this experimental section, the results show that the proposed ECPRC system works optimally compared to the other systems.

Index Terms—Agriculture, crop and leaf diseases, ensemble support vector machine (SVM), Internet of Things (IoT).

I. INTRODUCTION

ENSEMBLE classification and pattern recognition for crop monitoring system (ECPRC) for plant disease recognition supports the farmer to take proactive decisions at the beginning point of the disease growth. Manual or visual recognition

of these diseases from the observed plant portions (patterns) is inaccurate and inefficient. Moreover, the way of recognition requires a well-talented botanist. The issues that occur during conventional approaches can be rectified using ECPRC automatically.

ECPRC focuses on the development of spontaneous recognition and classification of the crop diseases from the gathered high-resolution multispectral images, hyperspectral images, and stereotype images. For example, sugar beet leaves are usually infected by the diseases like Downy mildew, rust, Cercospora leaf spot, and phoma leaf spot. In this regard, Rao and Sridhar [1] and Sreekantha and Kavya [2] analyzed the intelligent crop monitoring and irrigation system to help in the growth of the plants. Shaikh *et al.* [15] analyzed the green technologies applications in the agriculture fields.

Ray [3] and Tzounis *et al.* [7] explained the Internet of Things (IoT) future trends and architectures, respectively. IoT devices include high-resolution spectral cameras, agriculture sensors, and a Raspberry Pi controller which has been manipulating the plant images using machine learning (ML) algorithms. The images taken by the cameras are processed by extracting the leaf portions of the images.

Then, the extracted leaf regions are segmented using Otsu's method. Once the leaf regions are segmented, the disease portions are identified by using the Sobel edge detection approach for pointing disease-infected points in the leaves. These image processing techniques are used to spot the disease in the plant leaves but not to classify them efficiently. In order to achieve disease pattern-wise classifications, and recognition of diseases, many ML approaches are taken by the researchers.

For example, agricultural researchers, Marjanin [4], focused on the problems and solutions of agriculture development. Further, the other researchers developed the techniques, such as support vector machine (SVM), PCA, convolutional neural network (CNN) [14], [17], [18], decision trees, backpropagation neural networks, and *K*-nearest neighbors' classification for the disease classification. On the way, Lee *et al.* [8] pointed out phytophthora infest diseases on the tomatoes. They used various IoT components to build the system. Likewise, Xu *et al.* [9] applied SVM cotton mold diseases based on their visual marks. The main scope of this proposed ECPRC scheme is to develop a versatile crop disease monitoring system and IoT-based automatic decision-making system.

Existing works mainly focus on offline or outdated data items for leaf disease detection. This is taken as a serious

Manuscript received August 14, 2020; revised January 18, 2021 and March 21, 2021; accepted April 2, 2021. Date of publication April 13, 2021; date of current version August 6, 2021. (*Corresponding author: Amir H. Gandomi.*)

Gayathri Nagasubramanian and Rakesh Kumar Sakthivel are with the School of Computing Science and Engineering, Galgotias University, Greater Noida 201307, India (e-mail: gayathrierm@gmail.com; rakesherme@gmail.com).

Rizwan Patan is with the Department of Computing Science and Engineering, Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada 520007, India (e-mail: prizwan5@gmail.com).

Muthuramalingam Sankayya is with the Department of Information Technology, Thiagarajar College of Engineering, Madurai 625006, India (e-mail: smrit@tce.edu).

Mahmoud Daneshmand is with the School of Business, Stevens Institute of Technology, Hoboken, NJ 07030 USA (e-mail: mahmoud.daneshmand@stevens.edu).

Amir H. Gandomi is with the Faculty of Engineering and IT, University of Technology Sydney, Sydney, NSW 2007, Australia (e-mail: gandomi@uts.edu.au).

Digital Object Identifier 10.1109/IIOT.2021.3072908

problem, which is the scope of the proposed work, that aims to provide real-time IoT data analysis. The other contribution of this work is to develop ML- and DL-based multiclassification models for detecting the real-time IoT (sensors) inputs gathered from various leaves. In addition, this work mainly concentrates on the efficient application of ensemble nonlinear SVM (ENSVM), K -nearest, and CNN in leaf disease detection. The comparative experimental analysis helps to analyze the leaf patterns and disease particles from different perspectives.

The overall article is organized as follows. Section II gives the literature review, Section III provides the proposed work, Section IV provides the results, and finally, Section V ends with the conclusion.

II. LITERATURE REVIEW

The agriculture field requires more attention to be developed with intelligent monitoring and decision-making systems. Different types of research were carried out to identify various crop diseases. Rao and Sridhar [1] used several strategies to precise the captured indications in the cotton leaves. The images used in this work were maintained at nominal resolutions. These images were preprocessed and located in a separate image database. The conventional edge detection and classification techniques were used by the authors to identify the cotton leaf diseases. In addition to that, K -nearest classification was used to extract the color features of different leaves and SVM was applied to identify the diseases in plant leaves efficiently.

Sreekantha and Kavya [2] projected a disease monitoring system for cotton plant diseases. The diseases were recognized by them using the images, that reflect the disease spot marks on the cotton leaves. The authors processed the images under commercial mobile cameras in an uncontrolled situation of the farmland. This type of image processing makes the leaf segmentation critical. This work used two different types of sequenced classifiers.

At the first stage, the classifier isolates the leaf from the image background using local image features. After that, the classifier spots out the diseases using the hue and luminance of the image.

This kind of research signs is good to apply ML techniques for the disease monitoring system in IoT-based farming fields. On this research path, Marjani [4] delivered a grayscale morphology-based leaf nerve extraction system. To extract the leaf features, the color images of the leaves were converted into grayscale images based on the intensity and hue contents of the image. Then, grayscale morphology technique was taken to remove a color confusing section of the image. The next step is used to identify the grayscale variations for the leaf part and nerve part. In the end, an optimal threshold value is determined to isolate the nerve part of the leaf from others. These works were mainly used to recognize the leaf features which are helpful to identify the disease portions at different points of any leaf.

Further, Rajeswari *et al.* [5] identified the applications of X-ray image processing to spot the plant diseases. In this work, the leaf's 2-D images were used for extracting the geometrical features and digital morphological features. The goal

of this research work was to familiarize the appropriate features from the leaf patterns for future agricultural research. Mahdaveinejad *et al.* [6] proposed the ANN-based plant-type identification system from the given leaf images. In addition to the image processing techniques, the authors used the properties of the given leaf image (leaf aspect ratio, apex attributes, deviation ratio, and leaf angle). These leaf properties were given to ANN as a set of input. The authors collected around 550 leaf patterns for various plants. According to that, they trained more than 400 leaves and acquired 93% accuracy in the plant identification.

Srinivasalu *et al.* [13] used SVM for recognizing the cotton plant diseases, such as bacterial infections, *Cercospora*, and *Alternaria*. This work was extended in which, the results produced by SVM were given to the Android platform for better projection. This system makes the IoT platform work with ML techniques. There are many research works which have made their contributions in the field of the crop monitoring system. The proposed ECPRC concentrates on the sugar beets diseases, which is common in India. In this work, ENSVM (vital) is applied for the classification of the diseases. The conventional SVM has basic classification rules which are not effectively adaptable for multiple disease classification. However, ENSVM has multiple subclassifiers compared to the other techniques.

ML is a technology used to train the computer system for making decisions based on the learning techniques. ML techniques are generally used to give learning capability to the computer systems. This is called the skill to learn. There are various ML techniques available. They are:

- 1) supervised learning;
- 2) unsupervised learning;
- 3) clustering and classification;
- 4) data preprocessing;
- 5) dimensionality reduction;
- 6) multilayer perceptron;
- 7) nonparametric methods;
- 8) hidden Markov model.

Among these techniques, this proposed work uses an SVM with ensemble nonlinear classification. Unlike SVM (linear), ensemble nonlinear classification uses multilayer classifiers, which support uncertain conditions of leaf pattern detections. This helps to project more efficient classified results on disease detection. Similarly, CNN is applied for detecting leaf diseases. CNN is a type of multilayer perceptron model. Mainly, it is applied with many layers of filters. The filter types may be implemented as 5×5 , 4×4 , 3×3 , ... CNN does convolution at the beginning of filter sections to split the image patterns. Finally, it does deconvolution for getting the integrated results.

The multilayer perceptron is referred to the fully connected neural networks. This means each neuron of a particular layer is connected to all neurons in the succeeding layer. However, CNNs take the benefits of the hierarchical pattern in the leaf data and create more complex leaf patterns for generating meaningful complex data. This gives more accurate results in disease detection. Finally, this proposed work uses the K -means clustering technique for finding the classes of various disease patterns in a leaf. K -means clustering is a technique,

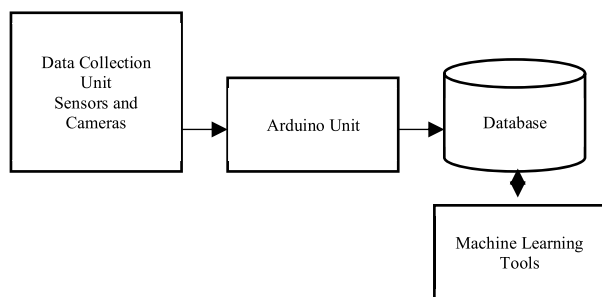


Fig. 1. Basic system architecture.

which uses the vector quantization approach. This is mainly derived from the signal processing technique. The technique is developed from the steps of cluster analysis in the data mining domain.

The K -means clustering technique was developed of the classification partitions with n number of continuous observations into k time of cluster units. Each observation gathered at different time intervals belongs to any one of the clusters. The clustered observations maintain the mean value. This is nonpolynomial (NP) hard in the computational aspect. This has a direct conceptual relationship with the K -nearest neighbor classification. The proposed system uses this technique for leaf disease classification.

III. PROPOSED APPROACH

In this proposed system, temperature sensors, soil moisture sensors, and light sensors are used to measure the temperature value, moisture level, and light intensity level of the plants, respectively. The data gathered from these sensors are manipulated by the Arduino UNO microprocessor unit. Fig. 1 provides the overall system architecture for the process of disease detection.

Then, the values are used by ENSVM and CNN to identify the plant disease. To achieve this disease analysis, ENSVM and CNN-based algorithms are used to make a training sample of leaf images and optimal threshold values for the leaf features. There are three modules in this proposed system. They are:

- 1) hardware module;
- 2) analysis module;
- 3) decision-making module.

The hardware module contains the Arduino controller unit, sensors, and cameras, and the analysis module maintains ML techniques activation. This module covered training and testing information about the leaves and leaf patterns. In the sensing module, the following sensors are used in the farming field. They are used to monitor each and every point in the plant. In addition to that, multiple sensors are deployed to measure environmental conditions. These components are connected via relays to the respective terminal devices, such as water motor and other computer systems. The design of the IoT architecture depends upon the components required to achieve the goal.

Ray [10] analyzed different types of IoT architectures in detail. The components involved in the IoT system include any

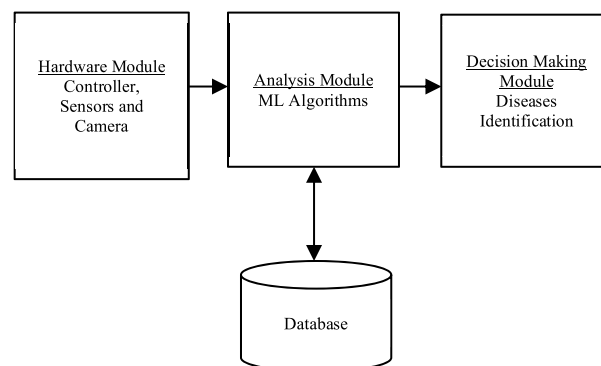


Fig. 2. ECPRC modules description.

TABLE I
IoT COMPONENTS

S.NO	Components-IoT	Model
1	Temperature sensor	LM 35 TO-92-3
2	Soil moisture sensor	LM358
3	Humidity sensor	DHT11
4	Light intensity sensor	BH1750
5	Hyper spectral cameras	HySpex
6	Water flow sensors	YF-S201
7	Microcontroller	Arduino UNO
8	Communication technologies	GSM module and WIFI

communicating device, terminal device, wireless sensor nodes, and networks. Ojha *et al.* [11] discussed the adaptation of the wireless sensor network in the IoT design. The represented components can communicate with the mobile phones to give the manipulated results regarding the disease classification, sensed environmental objects, and farmer suggestions. This kind of farming suggestion is given by the ML algorithms, SVM and CNN. Apart from these techniques, the component selection process is inevitable. Table I illustrates the details of the components used in this proposed work. Fig. 2 gives the specific pattern from the general structure in Fig. 1 to identify the plant disease.

A. ENSVM-Based Disease Identification

SVM is an ML technique which is used to analyze and classify the given problem in a supervised manner. The supervised classification using SVM uses different classes of disease patterns and leaf patterns to classify the observed plant diseases. This proposed ECPRC uses ENSVM. The ENSVM manages the data classification by dividing the classifiers into several subfunctions to spot out the disease variations. This is the advantage of detecting various leaf diseases. Zhang *et al.* [16] used ensemble SVM (ENSVM) for pattern identification for cognitive tasks identification. As same as, this proposed system as discussed in Algorithm 1, uses ENSVM for leaf disease pattern identification. The ENSVM classifies the input image patterns into several classes, c and subclasses, with n samples

Algorithm 1 ENSVM-Based Disease Detection**Input:** Dataset P_L , Classifier Splits, Samples m ;**Output:** Classified data instances**Step 1:** Split the image data into subsets $\{I_{kj}\}_{j=1}^S$ **Step 2:** Find sample disease instances, I_D **Step 3:** Acquire ENSVM AS_j from each subset I_{kj} **Step 4:** A representation of sample estimation $\in \mathbb{R}^{x \times s}$

$$M = \text{est}(\text{samp}, AS_n) \quad (4)$$

Step 5: Learn the EMSVM samples**Step 6:** Return the samples**Step 7:** Execute Smoothing function, SENSVM,

$$S(n, a) = n + \sum_{n=1}^{\infty} \frac{1}{B(\log(1 + Bn))} \quad (5)$$

B-log distribution

Step 8: Redo for all observed leaves, L_n .and k patterns

$$c \in \{-n, +n\} \text{ with respect to } c = \text{sign}(f(k)). \quad (1)$$

$f(k)$ is a nonlinear grouping of kernels $m(ki, k)$ calculating the resemblances between the training vector a and the given pattern vector, k , and ki . Here, ci represents the individual classifier unit in the multiclassifier module, where i varies from 1 to n . ai means the inputs of the individual classifier unit ci

$$f(k)^n = \sum_{k=0}^n a_{ci} \cdot m(ki, k)^{n-k}. \quad (2)$$

ENSVM supports multiple classifier units according to various leaf patterns. The classifiers $CS_1, \dots, CS_j$ use the production rules, $PR_1, \dots, PR_j$

$$PR_j(k)^n = \sum_{k=1}^n PR_j^{n-k}. \quad (3)$$

Fig. 3 illustrates the general process of ENSVM and the other classifiers in the ECPRC framework. In this figure, sensors and relays are shown in the hardware part. The ECPRC engine helps to activate either ENSVM or CNN activities once the data are received from the sensors. In addition, the overall units are controlled by the Arduino UNO module. The processed data can be stored anywhere as in cloud storage, local storage area, or remote database. This data is accessible from any place when it is hosted in the Internet domain. Preferably, they can be stored in portable cloud storage for providing security, accessibility, availability, and easy handling tasks. The ENSVM classifier handles the given testing data using the training data set. The training data set contains around 600 leaf patterns and 35 disease patterns.

The disease patterns may belong to several disease classes. The ENSVM is also used to identify the nutrients in each leaf. However, this can be analyzed by CNN when compared to ENSVM. Apart from disease identification, ENSVM helps to give accurate chlorophyll, potassium, and nitrogen levels in plants. As same as, Lee *et al.* [12] used various types of prediction systems for big data systems.

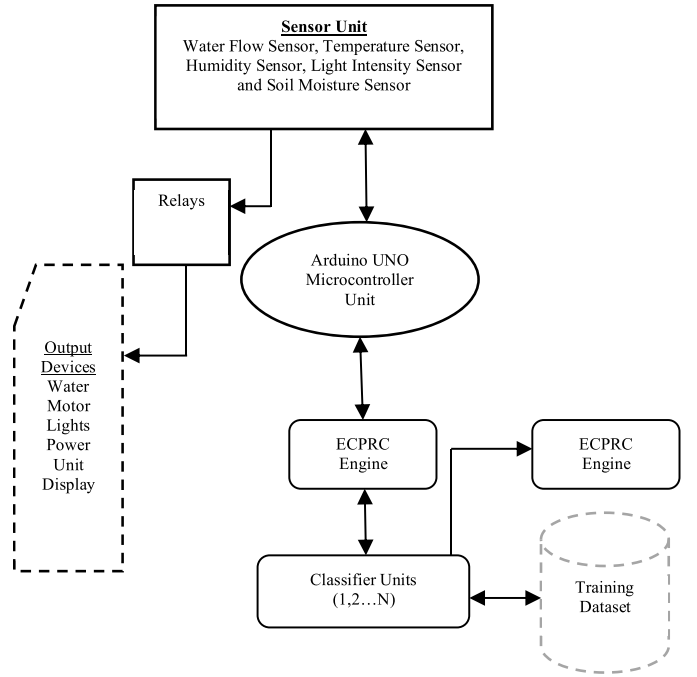


Fig. 3. ECPRC process for crop disease monitoring.

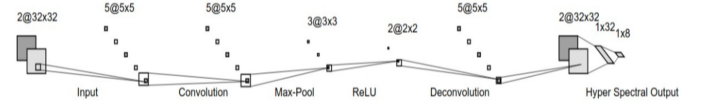


Fig. 4. CNN structure.

B. CNN-Based Disease Detection

CNN has three different layers, such as the input layer, the output layer, and the hidden layer as same all the neural networks as depicted in Fig 4. CNN holds multiple stages of functions as follows:

- 1) input phase;
- 2) convolution phase;
- 3) detector phase;
- 4) pooling phase;
- 5) normalization phase;
- 6) output phase.

CNN reduces the neural connections and complexity of the network functions to increase performance execution. Algorithm 2 describes CNN-based crop disease detection. Dahikar and Rode [14] used CNN for predicting and recognizing the crop yield in the agricultural lands. This motivates the proposed system to use CNN for disease prediction in the leaves. Compared to SVM and ENSVM, CNN uses trained neuron functions and layers of filters for isolating different disease sectors with high-quality rate. The following algorithm is proposed using CNN which detects the diseases based on filtering applied over spectral features of the leaf images. This works better than ENSVM's subclassifiers.

The main aim is to utilize the metaclassifier activities of ENSVM and CNN for detecting multiple leaf diseases. The motivation of this work lies in the usage of ML-based trained classifiers units of both ENSVM and CNN (DL) on dot spots identifications. These classifiers help to categorize multiple dot spots under multiple classes, which is an efficient way

Algorithm 2 CNN-Based Disease Detection**Input:** Image dataset, Samples**Output:** Disease Detections**Step 1:** Input image I_m (raw image) has $32 \times 32 \times 3$ size with RGB band (spectral bands)**Step 2:** Each image segments are weighted by convolutional layer function and form the neurons**Step 3:** Image filters are applied (15 Filters)**Step 4:** Relu activation function is applied on leaf elements with respect to a computed threshold value.**Step 5:** The down sampling operations have been activated using the pooling procedure using spatial details, $SP_1, \dots, SP_m$.**Step 6:** Compute classes and scores for different leaves and create CIFAR-10 set.**Step 7:** Create score-based disease symptoms in each leaf.**Step 8:** Redo the operationsTABLE II
CNN PHASES

Layers	Function	Filter Sizes	Channels	Filters
1	Convolution Pooling	5x5 3x3	6	56
2	Convolution Pooling	3x3 2x2	56	256
3	Convolution	3x3	256	256
4	Convolution	3x3	256	256
5	Convolution	3x3	256	256

of detecting the disease classification. The main advantage of using ENSVM is providing subclassifiers for various element properties. But this is limited in ML aspects. In advance, CNN is used with multifiltering components (DL).

The image segmentation, edge detection, and denoising processes of the crop disease detection system are implemented using the MATLAB tool. The Android studio is used to develop an application in mobile phone for successful monitoring of diseases and decision-making processes. In addition to these tools, the microprocessor uses an inbuilt operating system that governs the data gathered from different sensors. In CNN, $m \times m$ neural network layers are followed by a convolutional layer. For filter F_s , with $f \times f$, dimension, the output of the convolutional layer is, $(m - f + 1) \times (m - f + 1)$. Here, the prenonlinearity is denoted as $p^{\ell}jk$. The weighted filter components are computed for preceding neural layers as

$$p^{\ell}jk = \sum_{s=0}^m \cdot \sum_{l=0}^m \binom{n}{k} F_s l x^{k-1} (j+s)(k+l)^{n-k}. \quad (6)$$

Then, CNN uses a nonlinear function

$$x^{\ell}jk = \sigma(x^{\ell}jk). \quad (7)$$

Here, σ —nonlinear function, and $x^{\ell}jk$ —filters applied for input leaf sequences. The convolution and pooling phases are designed using 256 filters maximum as given in Table II. Further, Table II illustrates the details of the neural layers, filter sizes, and channels used to reconstruct the spectral images of the leaves at high-intensity levels.

IV. RESULTS

The input image of a leaf is given with RGB spectral information. The high-spectral image taken by HySpexHD



Fig. 5. Input image patterns. (a) Given RGB image. (b) Grayscale image.

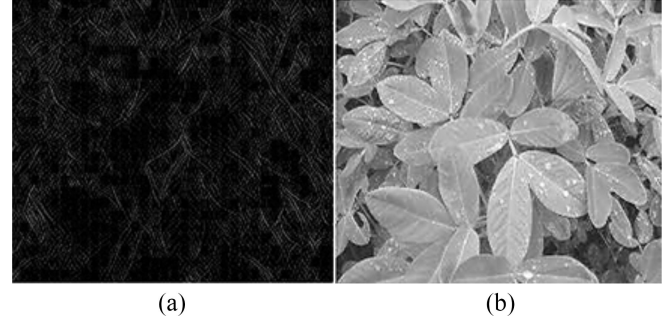


Fig. 6. Image recreation. (a) Noise coefficients. (b) Filtered output.

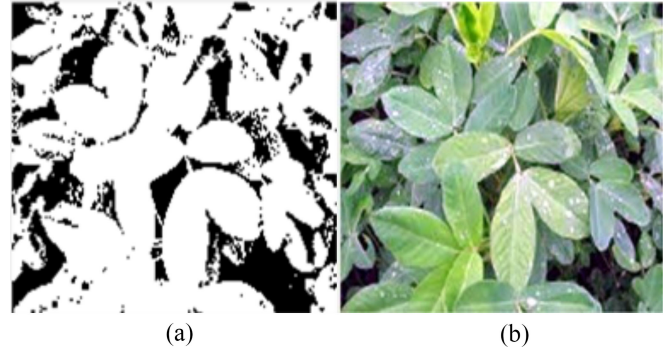


Fig. 7. Image enhancement. (a) Binary image. (b) Color-enhanced image.

camera is converted into its equivalent grayscale image. 75% of the data from the data set are used for training purposes and the remaining 25% are used for testing purposes.

Fig. 5(a) illustrates an RGB spectral image of the leaves and Fig. 5(b) depicts a grayscale image of the leaves. In the grayscale image, the dots in leaves are spotted clearly than the basic spectral view. Fig. 6(a) shows the noise coefficients of a given image, which gives unwanted depiction in an image. This phase helps to identify the leaves alone in the image. Further, these noise coefficients are used for denoising the image.

In addition, Fig. 6(b) gives another outcome, which filtered the leaf images. This filters the noise data. By comparing the training sets of these kinds of leaves, ENSVM and CNN identify the disease type. Fig. 7(a) provides a binary image to identify the exact leaves portion in the image. Then, the color enhancement helps to improve the contrast of the image to increase the visual interpretation of the image [Fig. 7(b)].

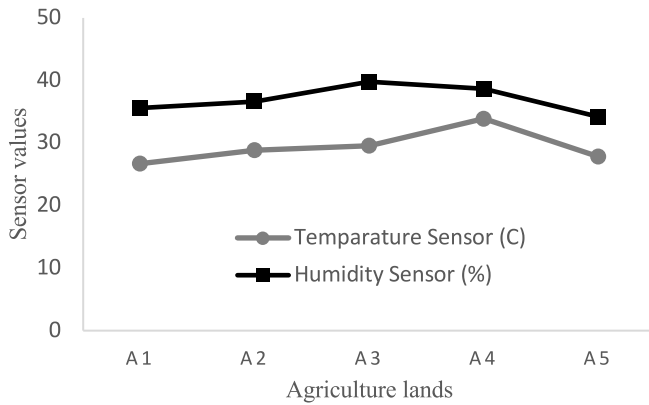


Fig. 8. Sample sensor readings.

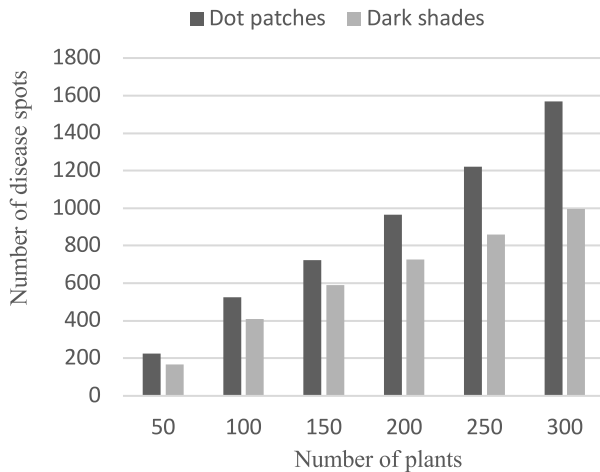


Fig. 9. Defects identification.

The Arduino UNO controller and the sensor units are illustrated with the display modules. Fig. 8 gives the reading of the temperature sensor (Celsius) and the humidity sensor (%) units with respect to the different croplands. The readings of the sensor vary from field to field. This kind of assistantship helps to clarify the environmental issues of the crop disease monitoring system.

Be sure that the symbols in your equation have been defined. Fig. 8 represents the level of temperature and humidity in different agriculture fields. The temperature and the humidity of the field are linear to each other. Fig. 9 illustrates the defects in plant leaves (dark shades and dot spots).

True positive and false negative are accurate decisions. This means true positive refers to the detection of disease and false-negative states the successful detection of a disease-less leaf. In contrast, true negative illustrates the disease where the leaf has no disease.

As same as, false-positive gives a false alarm for disease detection. The variations of CNN and ENSVM with respect to their accuracy and recall rate, respectively. This shows that CNN gives better results than ENSVM. The *K*-nearest gives a moderate classification rate compared to the other two proposed techniques. In this manner, Table III provides a comparison between CNN, ENSVM, and *K*-nearest classification techniques. The techniques, such as Naïve Bayes,

TABLE III
QUALITY MEASUREMENTS

Techniques	Performance Metrics (%)			
	Recall	Specificity	Accuracy	precision
ENSVM	80.6	79.5	75	78.7
CNN	85.6	83.9	80.1	84.6
k-nearest	80.2	78.7	74.6	76.4

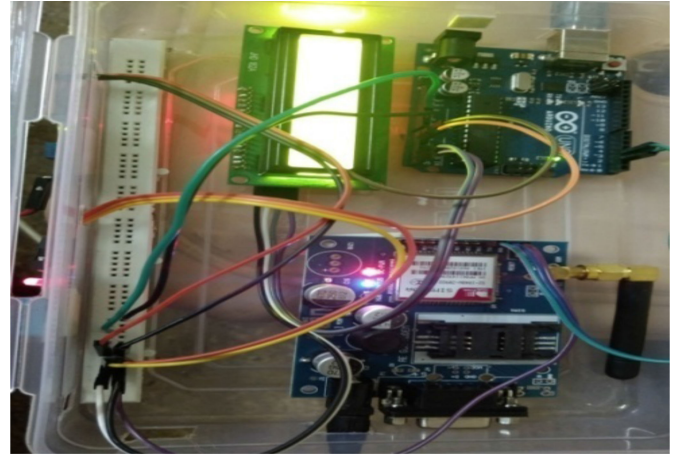


Fig. 10. Connection of monitoring system.

K-nearest, and other conventional approaches, are compared to the CNN and ENSVM, where the later approaches use multiple subclassification layers and spectral features-based filters. Moreover, these techniques are not more complex to create overhead in tiny IoT devices. It is very important to detect diseases from leaves using high-level spectral features and more subclassifiers than more complexity.

The results are compared with the existing techniques [16] to evaluate the proposed system performance. The existing system uses ENSVM and Gaussian models for pattern identification with the task evaluation. In their models, they used a limited set of patterns and applied ML techniques only. But the proposed system deals with both ML and DL techniques. In addition, most of the techniques consider offline data sets. But the proposed system takes the real-time sensor data to gain real efficiency.

Fig. 10 represents the connections of the agriculture monitoring system, temperature sensor, soil moisture sensor, and light intensity sensor. They are connected and the signals are amplified and sent for manipulating to the Arduino controller unit. LCD is used to display the values from the sensors. GSM is used to send the alert message to the farmer.

Fig. 11 provides the various quality measurements in terms of recall, sensitivity, specificity, and precision for various techniques. Figs. 12 and 13 describe the performance and the comparative analysis of various ML techniques and ENSVM technique. Fig. 12 illustrates the absolute error rate in leaf pattern recognition. Fig. 13 depicts the computation overhead created by each ML and DL technique. In this comparison, CNN and ENSVM perform notably. Particularly, ENSVM effectively detects the leaf patterns. At the same time, CNN detects the leaf patterns and disease spots with minimal errors.

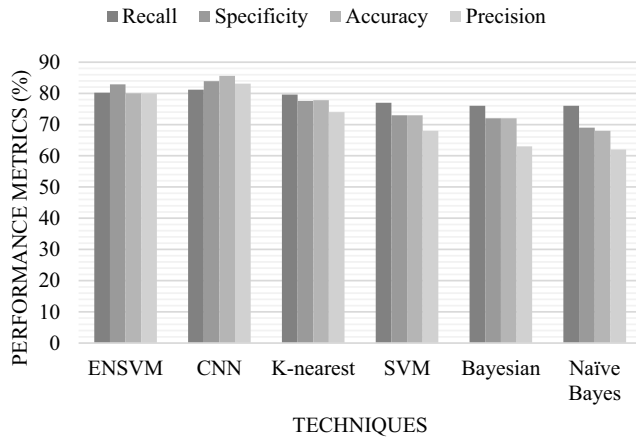


Fig. 11. Quality measurements.

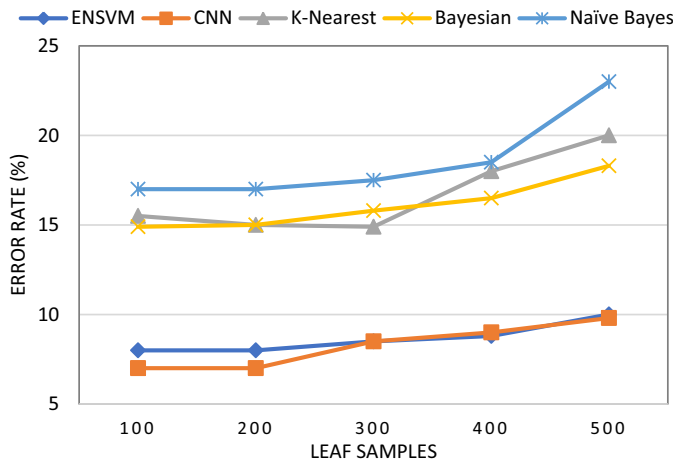


Fig. 12. Error rate.

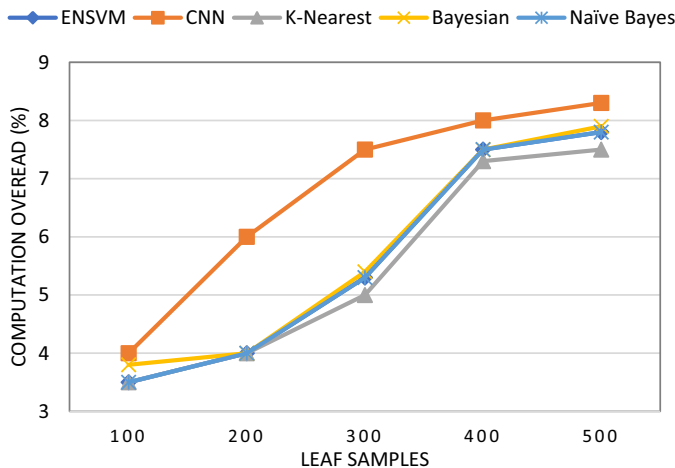


Fig. 13. Computation overhead.

The reason is that CNN is mainly implemented for image pattern recognition purposes.

Fig. 13 reveals the computation overhead of all the techniques. In this comparison, CNN produces little more overhead than ENSVM and other techniques. As CNN is a DL network, it has more complex filter functions than other systems. But the need for efficiency is important in disease identification. Thus, the overhead can be ignored. Other techniques, such as

K -nearest, SVM, Bayesian network, and Naïve Bayes, produce limited overhead. However, they lack in accuracy rate. The evaluation process is experimented for increasing the number of leaf samples.

V. CONCLUSION

The ECPRC system was implemented to provide both environmental observance-based crop disease monitoring system. In this work, IoT infrastructures were deployed to monitor and capture the leaf images in hyperspectral modes. The hyperspectral images of the leaves were processed to track the disease points. These images were stored in cloud storage systems to be handled by ENSVM and CNN. CNN and ENSVM were used to train the IoT infrastructures to detect the various types of leaf diseases. In addition, the other ML techniques, such as K -nearest, Bayesian, Naïve Bayes, and conventional SVM techniques, were compared with each other for ensuring the real-time application of the intelligent crop disease monitoring system. In the future, this work is to be extended using a huge data set of multiple plants and its parts using the deep learning approaches.

REFERENCES

- [1] R. N. Rao and B. Sridhar, "IoT based smart crop-field monitoring and automation irrigation system," in *Proc. 2nd Int. Conf. Inventive Syst. Control (ICISC)*, 2018, pp. 478–483.
- [2] D. K. Sreekantha and A. M. Kavya, "Agricultural crop monitoring using IOT-a study," in *Proc. 11th Int. Conf. Intell. Syst. Control (ISCO)*, 2017, pp. 134–139.
- [3] P. P. Ray, "Internet of Things for smart agriculture: Technologies, practices and future direction," *J. Ambient Intell. Smart Environ.*, vol. 9, no. 4, pp. 395–420, 2017.
- [4] M. Marjani, "Big IoT data analytics: Architecture, opportunities, and open research challenges," *IEEE Access*, vol. 5, pp. 5247–5261, 2017.
- [5] S. Rajeswari, K. Suthendran, and K. Rajakumar, "A smart agricultural model by integrating IoT, mobile and cloud-based big data analytics," in *Proc. Int. Conf. Intell. Comput. Control (I2C2)*, 2017, pp. 1–5.
- [6] M. S. Mahdavinjad, M. Rezvan, M. Barekatain, P. Adibi, P. Barnaghi, and A. P. Sheth, "Machine learning for Internet of Things data analysis: A survey," *Digit. Commun. Netw.*, vol. 4, no. 3, pp. 161–175, 2018.
- [7] A. Tzounis, N. Katsoulas, T. Bartzanas, and C. Kittas, "Internet of Things in agriculture, recent advances and future challenges," *Biosyst. Eng.*, vol. 164, pp. 31–48, Dec. 2017.
- [8] M.-H. Lee, J.-H. Hwang, and H. Yoe, "Agricultural production system based on IoT," in *Proc. IEEE 16th Int. Conf. Comput. Sci. Eng.*, 2013, pp. 833–837.
- [9] J. Xu, J. Zhang, X. Zheng, X. Wei, and J. Han, "Wireless sensors in farmland environmental monitoring," in *Proc. Int. Conf. Cyber-Enabled Distrib. Comput. Knowl. Disc.*, 2015, pp. 372–379.
- [10] P. P. Ray, "A survey on Internet of Things architectures," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 30, no. 3, pp. 291–319, 2018.
- [11] T. Ojha, S. Misra, and N. S. Raghuvanshi, "Wireless sensor networks for agriculture: The state-of-the-art in practice and future challenges," *Comput. Electron. Agric.*, vol. 118, pp. 66–84, Oct. 2015.
- [12] J. Lee, E. Lapira, B. Bagheri, and H.-A. Kao, "Recent advances and trends in predictive manufacturing systems in big data environment," *Manuf. Lett.*, vol. 1, no. 1, pp. 38–41, 2013.
- [13] P. Srinivasulu, M. S. Babu, R. Venkat, and K. Rajesh, "Cloud service-oriented architecture (CSOA) for agriculture through Internet of Things (IoT) and big data," in *Proc. IEEE Int. Conf. Electr. Instrum. Commun. Eng. (ICEICE)*, 2017, pp. 1–6.
- [14] S. S. Dahikar and S. V. Rode, "Agricultural crop yield prediction using artificial neural network approach," *Proc. Int. J. Innovat. Res. Electr. Electron. Instrum. Control Eng.*, vol. 2, no. 1, pp. 683–686, 2014.
- [15] F. K. Shaikh, S. Zeadally, and E. Exposito, "Enabling technologies for green Internet of Things," *IEEE Syst. J.*, vol. 11, no. 2, pp. 983–994, 2015.
- [16] J. Zhang, Z. Yin, and R. Wang, "Pattern classification of instantaneous cognitive task-load through GMM clustering, Laplacian eigenmap, and ensemble SVMs," *IEEE/ACM Trans. Comput. Biol. Bioinform.*, vol. 14, no. 4, pp. 947–965, Jul./Aug. 2017.

- [17] P. Samuel, S. Subbaiyan, B. Balusamy, S. Doraikannan, and A. H. Gandomi, "A technical survey on intelligent optimization grouping algorithms for finite state automata in deep packet inspection," *Arch. Comput. Methods Eng.*, vol. 28, pp. 1–26, Apr. 2020. [Online]. Available: <https://doi.org/10.1007/s11831-020-09419-z>
- [18] A. Telikani, A. H. Gandomi, A. Shahbahrani, and M. N. Dehkordi, "Privacy-preserving in association rule mining using an improved discrete binary artificial bee colony," *Expert Syst. Appl.*, vol. 144, Apr. 2020, Art. no. 113097.
- [19] G. Kasinathan, S. Jayakumar, A. H. Gandomi, M. Ramachandran, R. Patan, and S. J. Fong, "Automated 3-D lung tumor detection and classification by an active contour model and CNN classifier," *Expert Syst. Appl.*, vol. 134, pp. 112–119, Nov. 2019.



Gayathri Nagasubramanian received the M.E. degree in computer science and engineering from Thiagarajar College of Engineering, Madurai, India, in 2016.

She is currently working as an Assistant Professor with the School of Computing Science and Engineering, Galgotias University, Greater Noida, India. She has published reputed 25 (SCI journals and Scopus indexed journals), and also presented papers in national/international conferences, published book chapters in CRC Press, IGI Global,

Springer, Elsevier, and also have edited books in Wiley and CRC Press. She has seven Indian patents, one Australian patent. She is a Guest Editor for *Recent Patents on Computer Science*. Her research interests include bigdata analytics, IoT, and networks.

Prof. Nagasubramanian is an Active Reviewer in *Bentham Science Journal*, *Mobile Networks and Applications*, *Expert Systems With Applications*, *Sustainable Computing: Informatics and Systems*, *IEEE INTERNET OF THINGS*, and *Expert Systems*.



Rakesh Kumar Sakthivel received the B.Tech. degree in computer science and engineering and the M.E. degree in computer science and engineering from Anna University, Chennai, India, in 2012 and 2016, respectively.

He is currently working as an Assistant Professor with the School of Computing Science and Engineering, Galgotias University, Greater Noida, India. He has published reputed 20 (SCI journals and Scopus indexed journals), and also presented papers in national/international conferences, published book

chapters in CRC Press, IGI Global, Springer, Elsevier, and also have edited books in Wiley and CRC Press. He has six Indian patents. His research interests include bigdata analytics, IoT, and networks and cloud computing.

Dr. Sakthivel is an active reviewer for *Bentham Science*, *Mobile Networks and Applications*, *Expert Systems With Applications*, *Sustainable Computing: Informatics and Systems*, *IEEE INTERNET OF THINGS JOURNAL*, and *Expert Systems*.



Rizwan Patan (Member, IEEE) received the B.Tech. and M.Tech. degrees from Jawaharlal Nehru Technological University Anantapur, Anantapur, India, in 2012 and 2014, respectively, and the Ph.D. degree in computer science and engineering from the School of Computer Science and Engineering, VIT University, Vellore, India, in 2017.

He has been an Assistant Professor with the Department of Computer Science and Engineering, Velagapudi Ramakrishna Siddhartha Engineering College, Vijayawada, India, since 2019. From 2017

to 2019, he was the former Assistant Professor with the School of Computing Science and Engineering, Galgotias University, Greater Noida, India. He has published reputed 20 SCI journals and ten free Scopus indexed journals, and also presented papers in national/international conferences, published book chapters in CRC Press, IGI Global, Elsevier, and edited as books. He has ten Indian patents one U.S. patent.

Dr. Patan received the Award from World Research Council and American Medical Council in the Title of Innovative Researcher on Big Data and IoT in 2019. He is a Guest Editor for *International Journal of Grid and Utility Computing* (Inderscience), *Recent Patents on Computer Science*, *Information Medical Unlock* (Elsevier), and *Neural Computing and Applications* (Springer).



Muthuramalingam Sankayya received the Bachelor of Engineering degree in computer science and engineering from Bharathidasan University, Tiruchirappalli, India, in 1997, the Master of Engineering degree in digital communication and network engineering from Kamaraj University, Madurai, India, in 2002, and the Ph.D. degree in information and communication engineering from Anna University, Chennai, India, in 2012.

He is currently working as an Associate Professor with the Department of Information Technology, Thiagarajar College of Engineering, Madurai. He has published various quality papers in the reputed journals. His main thrust research areas are wireless network, mobile communication, ad hoc networks.



Mahmoud Daneshmand (Life Senior Member, IEEE) received the B.S. and M.S. degrees in mathematics from the University of Tehran, Tehran, Iran, in 1964 and 1966, respectively, and the M.S. and Ph.D. degrees in statistics from the University of California at Berkeley, Berkeley, CA, USA, in 1973 and 1976, respectively.

He is a Co-Founder and a Professor with the Department of Business Intelligence and Analytics and a Professor with the Department of Computer Science, Stevens Institute of Technology, Hoboken, NJ, USA. He has over 40 years of industry and university experience as a Professor, a Researcher, an Assistant Chief Scientist, an Executive Director, a Distinguished Member of Technical Staff, a Technology Leader, a Chairman of Department, and the Dean of School with Bell Laboratories, Murray Hill, NJ, USA; AT&T Shannon Labs—Research, Florham Park, NJ, USA; the University of California at Berkeley; the University of Texas at Austin, Austin, TX, USA; the Sharif University of Technology, Tehran; the University of Tehran; New York University, New York, NY, USA; and the Stevens Institute of Technology.

Prof. Daneshmand is the Steering Committee Chair of *IEEE INTERNET OF THINGS JOURNAL*.



Amir H. Gandomi (Senior Member, IEEE) received the Ph.D. degree in engineering from the University of Akron, Akron, OH, USA, in 2015.

He is a Professor of Data Science and an ARC DECRA Fellow with the Faculty of Engineering and Information Technology, University of Technology Sydney, Sydney, NSW, Australia. He was an Assistant Professor with the School of Business, Stevens Institute of Technology, Hoboken, NJ, USA, and a Distinguished Research Fellow with the BEACON Center, Michigan State University, East

Lansing, MI, USA. He has published over 200 journal papers and seven books which collectively have been cited 20 000 + times (H-index = 66). His research interests are global optimization and (big) data analytics using machine learning and evolutionary computations in particular.

Prof. Gandomi has been named as one of the most influential scientific mind and Highly Cited Researcher (top 1% publications and 0.1% researchers) for four consecutive years, 2017–2020. He also ranked 18th in GP bibliography among more than 12 000 researchers. He has served as associate editor, editor, and guest editor in several prestigious journals, such as an Associate Editor of *IEEE TRANSACTIONS ON BIG DATA* and *IEEE INTERNET OF THINGS JOURNAL*. He is active in delivering keynotes and invited talks.